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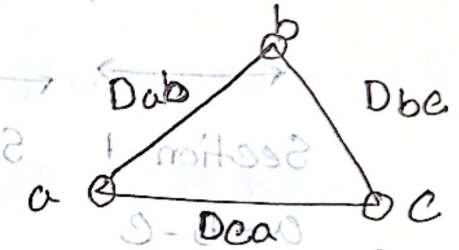
Ank- 2.3

Inductance of 3ϕ Transmission Line with Unsymmetrical Spacing

$$\lambda_a = 0.4605\left[I_a \log\left(\frac{1}{r'}\right) + I_b \log\left(\frac{1}{D_{ab}}\right) + I_c \log\left(\frac{1}{D_{ca}}\right)\right] \quad -(1)$$

$$\lambda_b = 0.4605\left[I_b \log\left(\frac{1}{D_{ab}}\right) + I_b \log\left(\frac{1}{r'}\right) + I_c \log\left(\frac{1}{D_{bc}}\right)\right] \quad -(2)$$

$$\lambda_c = 0.4605\left[I_c \log\left(\frac{1}{D_{ca}}\right) + I_b \log\left(\frac{1}{D_{bc}}\right) + I_c \log\left(\frac{1}{r'}\right)\right] \quad -(3)$$



$$\begin{bmatrix}\lambda_a \\ \lambda_b \\ \lambda_c\end{bmatrix} = 0.4605 \begin{bmatrix} \log(1/r') & \log(1/D_{ab}) & \log(1/D_{ca}) \\ \log(1/D_{ab}) & \log(1/r') & \log(1/D_{bc}) \\ \log(1/D_{ca}) & \log(1/D_{bc}) & \log(1/r') \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} \quad -(4)$$

$$L = 0.4605 \begin{bmatrix} \log(1/r') & \log(1/D_{ab}) & \log(1/D_{ca}) \\ \log(1/D_{ab}) & \log(1/r') & \log(1/D_{bc}) \\ \log(1/D_{ca}) & \log(1/D_{bc}) & \log(1/r') \end{bmatrix} \text{ mH/km} \quad -(5)$$

For balanced 3ϕ currents with I_a as reference;

(4) and (5)

I_a/I_a $\quad$ I_b/I_a

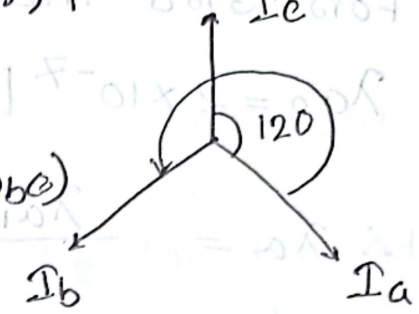
$$L_a = \frac{\lambda_a}{I_a} = 0.4605\left[\log(1/r') + \log(1/D_{ab})\frac{I_b}{I_a} + \log(1/D_{ca})\frac{I_c}{I_a}\right]$$

$$= 0.4605\left[\log(1/r') + \log(1/D_{ab})\,\alpha^2 + \log(1/D_{ca})\,\alpha\right] \quad -(7)$$

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$$L_b = 0.4605 \left[\log (1/D_{ab}) \alpha + \log (1/r') + \log (1/D_{bc}) \alpha^2 \right] \quad -(8)$$

$$L_c = 0.4605 \left[\log (1/D_{ca}) \alpha^2 + \log (1/D_{bc}) \alpha + \log (1/r') \right] \quad -(9)$$



7, 8, 9 → phase inductances are not equal.

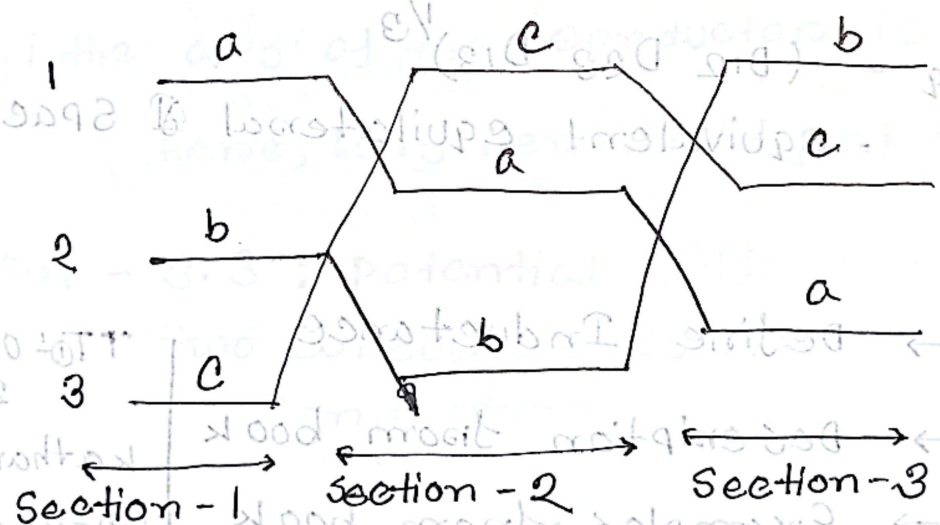
To make it same we use Transpose transmission line

$$\left.\begin{aligned} I_b &= \alpha^2 I_a \\ I_c &= \alpha I_a \end{aligned}\right\} \quad -(6)$$

Here, $\alpha = 1 \angle 120^\circ$

$\alpha^2 = 1 \angle 240^\circ$

~~Inductance~~



For the first inductor,

$$\lambda_{a1} = 2 \times 10^{-7} \left[I_a \ln (1/r'_a) + I_b \ln (1/D_{12}) + I_c \ln (1/D_{13}) \right]$$

For 2nd Inductor

$$\lambda_{a2} = 2 \times 10^{-7} \left[I_a \ln (1/r'_a) + I_b \ln (1/D_{23}) + I_c \ln (1/D_{12}) \right.$$

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For 3rd inductor,

$$\lambda_{a3} = 2\times10^{-7}\left[I_a \ln(1/r_a') + I_b \ln(1/D_{13}) + I_c \ln(1/D_{23})\right.$$

$$\therefore \lambda_a = \frac{\lambda_{a1} + \lambda_{a2} + \lambda_{a3}}{3}$$

$$= 2\times10^{-7}\left[I_a \ln(1/r_a') + I_b \ln\left\{\frac{1}{(D_{12} D_{23} D_{13})^{1/3}}\right\}\right.$$

$$\left. + I_c \ln\left\{\frac{1}{(D_{12} D_{23} D_{13})^{1/3}}\right\}\right]$$

But, $I_b + I_c = -I_a$

$$L_a = 2\times10^{-7} \ln\left(\frac{D_{eq}}{r_a'}\right)$$

$$L_a = L_b = L_c$$

Here, $D_{eq} = (D_{12} D_{23} D_{13})^{1/3}$

= equivalent equilateral Spacing

For Exam:

Art 2.2 → Define Inductance

Art 2.5 → Description from book

Art 2.7 → Examples from book

• Derivation, Example, Exercise

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• Skin Effect, Proximity Effect → Art 2.12

• Bundle conductor → Art 2.10

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Chapter - 03 : Capacitance of Transmission Lines

Art - 3.2 : Electric field of a long straight line Conductor

Consider two points P_1 and P_2 located at distances D_1 and D_2 respectively from the Conductor axis. The potential difference V_{12} (between P_1 and P_2) is given by,

$$V_{12} = \int_{D_1}^{D_2} \frac{q}{2\pi k x}\, dx = \frac{q}{2\pi k} \ln\left(\frac{D_2}{D_1}\right) V$$

As, Electric field intensity at a distance from the axis of the conductor is $E = \frac{q}{2\pi k x}$ V/m

Here, k is permitivity of medium

Art - 3.3 : potential difference between two conductors of a group of parallel conductors

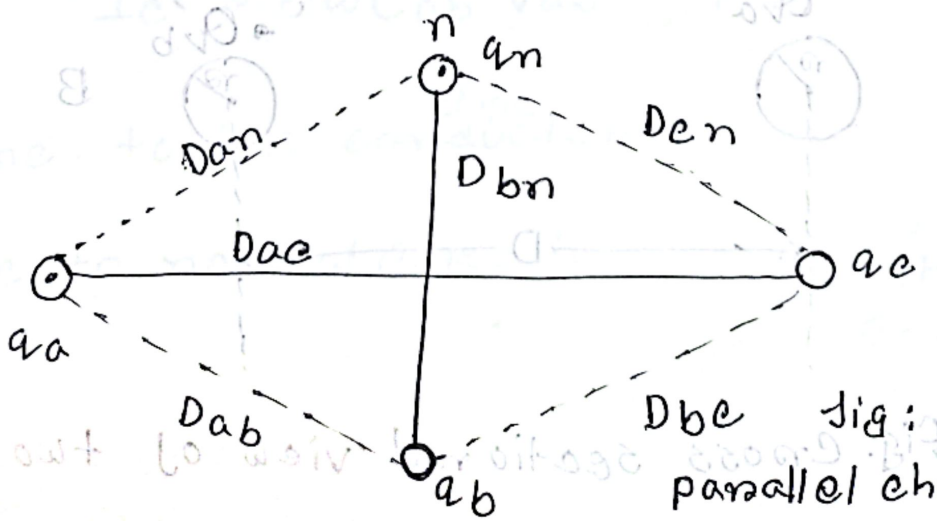


fig : A grop of parallel charged Conducton

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Assumptions :

→ Conductors are far removed from the group and sufficiently removed from each other conductors radii is much smaller than the distance between them

→ Charge on each conductor remains uniformly distributed around its periphery and length.

The potential difference between any two conductors of the group can be obtained by,

$$V_{ab} = \frac{1}{2\pi k}\left[q_a \ln \frac{D_{ab}}{r_a} + q_b \ln\left(\frac{r_b}{D_{ba}}\right) + q_c \ln\left(\frac{D_{cb}}{D_{ca}}\right) + \dots + q_n \ln\left(\frac{D_{nb}}{D_{na}}\right)\right] V$$

(1st, 2nd)

Art - 3.4 : Capacitance of a two wire line

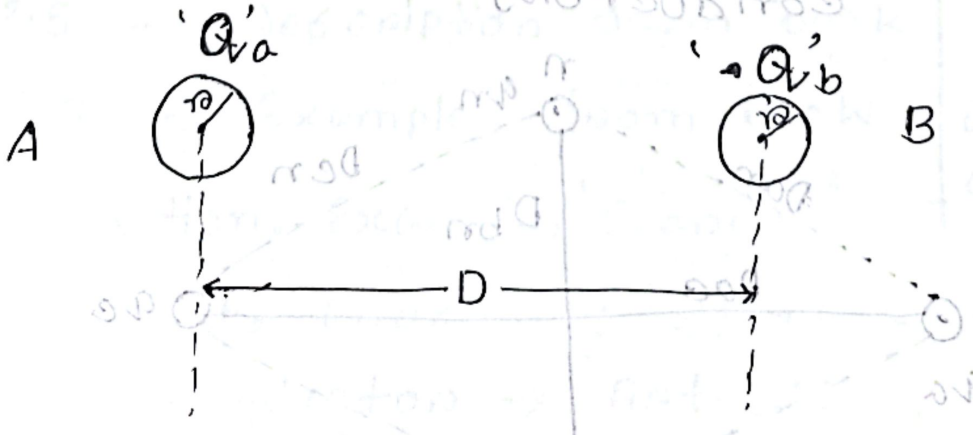


Fig Cross sectional view of two wire line

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The line develops equal and opposite sinusoidal charges on the two conductors which can be presented as phasors q_a and q_b so that

$$q_a = -q_b$$

So, $$V_{ab} = \frac{1}{2\pi k}\left[q_a \ln\left(\frac{D}{r_a}\right) + q_b \ln\left(\frac{r_b}{D}\right)\right]$$

Since, $q_a = -q_b$, $$V_{ab} = \frac{q_a}{2\pi k} \ln\left(\frac{D^2}{r_a r_b}\right)$$

The line capacitance, $$C_{ab} = \frac{q_a}{V_{ab}} = \frac{\pi k}{\ln\left(D/\sqrt{r_a r_b}\right)}$$

F/m length of line

$$= \frac{0.0121}{\log\left(D/\sqrt{r_a r_b}\right)} \mu F/km$$

If $r_a = r_b = r$

$$C_{ab} = \frac{0.0121}{\log(D/r)} \mu F/km$$

The associated line charging current

$$I_c = j\omega C_{ab} V_{ab} \quad V/km$$

a. Line to line capacitance : a ○—||—○ b (C_{ab})

b. Line to neutral capacitance : a ○—|| ($\frac{V_{ab}}{2}$) —n— || ($\frac{V_{ab}}{2}$) —○ b

$C_{an} = 2C_{ab}$ $C_{bn} = 2C_{ab}$

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$$C_n = C_{an} = C_{bn} = 2C_{ab}$$

$$= \frac{0.0242}{\log(D/r)} \quad \mu F/km$$

$$C_{an} = \frac{Q}{V_{an}} = Q / \frac{V_{ab}}{2}$$

$$= \frac{2Q}{V_{ab}}$$

Voltage divided into half

So, capacitance is increased.

eqn. 3.7

Art-3.5 ; Capacitance of a three phase line with equilateral spacing

3ϕ composed of three identical conductor of radius r in equilateral configuration

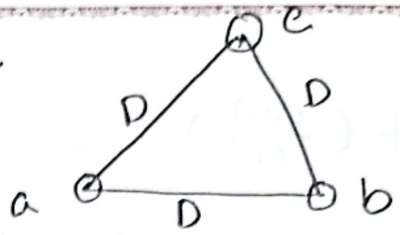
$$V_{ab} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\left(\frac{D}{r}\right) + q_b \ln\left(\frac{r}{D}\right) + q_c \ln\left(\frac{D}{D}\right)\right.$$

$$V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\left(\frac{D}{r}\right) + q_b \ln\left(\frac{D}{D}\right) + q_c \ln\left(\frac{r}{D}\right)\right]$$

$$V_{ab} + V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[2q_a \ln\left(\frac{D}{r}\right) + (q_b + q_c) \ln\left(\frac{D}{r}\right)\right]$$

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There are no other charge



$$q_a + q_b + q_c = 0$$

$$q_a + q_b = -q_c$$

$$V_{ab} + V_{ac} = \frac{1}{2\pi\varepsilon_0}\left[2q_a \ln(D/r) - q_a \ln(r/D)\right]$$

$$= \frac{1}{2\pi\varepsilon_0}\left[2q_a \ln(D/r) + q_a \ln(D/r)\right]$$

$$= \frac{1}{2\pi\varepsilon_0} 3q_a \ln(D/r)$$

$$V_{ab} + V_{ac} = 3V_a$$

$$3V_{an} = \frac{1}{2\pi\varepsilon_0} 3q_a \ln(D/r)$$

$$V_{an} = \frac{1}{2\pi\varepsilon_0} q_a \ln(D/r)$$

Capacitance of line to neutral

$$C_n = \frac{q_a}{V_{an}} = q_a \Big/ \frac{1}{2\pi\varepsilon_0} q_a \ln(D/r) = \frac{2\pi\varepsilon_0}{\ln(D/r)}$$

$$V_{ab} = V_{an} - V_{bn}$$

$$V_{ac} = V_{an} - V_{cn}$$

$$V_{ab} + V_{ac} = V_{an} - V_{bn} + V_{an} - V_{cn}$$

$$= 2V_{an} - V_{an} \angle -120^\circ$$

$$= V_{an} \angle -240^\circ = 3V_{an}$$

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